Yuhao Yuan

yuhao.yuan@mail.utoronto.ca +1 4376675335 Toronto https://yyhaos.github.io/

EDUCATION

University of Toronto **Sept 2025 – Jul 2027**

Master of Education

Beijing Normal University **Sept 2016 – Jul 2021**

Bachelor of Computer Science

ACADEMIC EXPERIENCE

Researcher - Dr. GPT **Jan 2026 - Now**

- Maintained the English and Chinese versions of an LLM-powered psychological assessment platform.
- Built CI/CD pipelines to automate build and deployment workflows.
- Contributed to platform deployment and service delivery in mainland China.

Researcher - pyKeyATM project **Apr 2025 - Now**

- Developed pyKeyATM, a Python and C++ package for keyword-assisted topic modeling, inspired by the R package keyATM.
- Implemented computational methods and research software components to support topic modeling workflows.

Researcher - rBKT Project **Aug 2024 - Dec 2025**

- Developed an open-source R implementation of Bayesian Knowledge Tracing for estimating student knowledge mastery from problem-solving sequences.
- Contributed to model training and prediction APIs, data validation, simulation, evaluation, parameter-recovery experiments, and package documentation.

Researcher - Fed IRT **Jan 2024 - Aug 2024**

- Co-developed FedIRT, an open-source R package for privacy-preserving estimation of 1PL, 2PL, and graded response models using federated learning.
- Implemented and evaluated FedAvg workflows and a Shiny client-server interface, enabling institutions to estimate IRT parameters collaboratively without sharing raw data.

Researcher - Entrepreneurial Project **Aug 2021 - Jul 2024**

- Developed an Android App called “keysManager” using Kotlin
- Launched the product outside of school, continuously tracked the users’ online activities, and updated the App according to the target market’s feedback
- Accomplished the goal of the App to manage user’s passwords, create secure passwords, set up network Keychain, and automatically sync all data while maintaining network security

Researcher - Graduation Design **Jan 2021 - Jul 2021**

- Developed methods for generating 3D model texture maps from actual images.
- An algorithm using convex hull nearest matching points to iteratively refine the projection matrix.
- Introduced an automated process for generating texture maps from multiple images, significantly reducing the need for manually marked initial correspondence points.

Researcher - Team Performance Evaluation Model in Soccer **Feb 2020 - Apr 2020**

- Collaborated on developing a comprehensive model to evaluate soccer team performance using network analysis of player interactions.
- Led the team and won the first prize.

PUBLICATIONS

- **Yuan, Y.**, Zhou, B., Qi, J., Luo, N., & Ji, F. (2026). BKT: A Bayesian knowledge tracing package for the R environment. *Behavior Research Methods, 58*(5), Article 134. doi:10.3758/s13428-026-02955-9
- Zhou, B., **Yuan, Y.**, Luo, N., Qi, J., & Ji, F. (2025). FedIRT: An R package and Shiny app for estimating federated item response theory models. *Journal of Open Source Software, 10*(113), 7869. doi:10.21105/joss.07869

CONFERENCE PRESENTATIONS

- **Yuan, Y.**, Zhou, B., Qi, J., Luo, N., & Ji, F. (2026). rbkt: An efficient R package for Bayesian knowledge tracing [Poster presentation]. *American Psychological Association Annual Convention*, Washington, DC, United States.

RESEARCH SOFTWARE

- **BKT** (2026). An R package for Bayesian knowledge tracing. CRAN
- **FedIRT** (2025). An R package and Shiny application for federated item response theory estimation. CRAN
- **pyKeyATM** (2026). A Python and C++ package for keyword-assisted topic modeling. PyPI

WORK EXPERIENCE

Data & Evaluation Intern – Vretta, Canada **Sept 2026 – Present**

- Contributed to the development of a TypeScript-based item response theory (IRT) scoring package for educational assessment.
- Implemented numerical quadrature methods to support expected a posteriori (EAP) ability estimation.
- Developed unit tests and benchmarks to evaluate numerical accuracy, computational performance, and reliability.

Software Developer - ByteDance Ltd. China **June 2021 - Sept 2024**

- Supported Continuous Integration and Continuous Delivery of basic components for China Tiktok iOS Application.
- Worked on solving user needs and making the app easier to use in different languages and regions.
- Developed two new applications as the main developer.

Frontend Software Developer Internship - Alibaba Ltd. China **Jun 2020 - Sept 2020**

- Used a fast technique called SIMD (Single Instruction, Multiple Data) to apply a blur effect and smooth out the pictures.

AWARDS

- National Competition Medalist Scholarship **2021**
- MCM/ICM (Mathematical Modeling) Finalist **2020**
- MCM/ICM (Mathematical Modeling) Meritorious **2018**
- International Collegiate Programming Contest (Shenyang) Silver Medal **2019**
- International Collegiate Programming Contest (Jiaozuo) Bronze Medal **2018**
- International Collegiate Programming Contest (Beijing) Bronze Medal **2017**
- China Collegiate Computing Contest 3rd Place (Group) **2019**
- China Collegiate Programming Contest (Harbin) Silver Medal **2019**
- China Collegiate Programming Contest (Qinhuangdao) Bronze Medal **2018**
- BNU Programming Contest (7th annual contest) 2nd Place **2018**
- BNU Programming Contest (6th annual contest) 2nd Prize **2017**
- Jingshi Academic Excellence Scholarship **2017**